

Nicolas Berkouk

Professional Experience

Since 2023	Scientific Lead, AI Department, CNIL, Paris • PANAME Project Manager: Spearheaded a cross-functional initiative of 30 researchers and developers across four institutions to build a unified open-source Python library for GDPR compliance and privacy-preserving AI evaluation. Defined the product roadmap, translated regulatory guidelines into technical requirements, and bridged the gap between a research consortium and an engineering team. • AI Research Lead: Formulated research roadmaps and coordinated cross-institutional projects on AI safety and privacy: – Model quantization and differential privacy (collaborating with INRIA). – Sociotechnical dimension of explainable AI and open-source AI development (with Sciences Po). – Representing CNIL in national cybersecurity initiatives (PEPR Cybersécurité) and with INESIA (the French AI Safety Institute). • AI Audit & Compliance Strategy: Designed CNIL's technical auditing framework for AI systems under the EU AI Act. • Technical Audits & Investigations: Led technical investigations and robustness controls on algorithmic and generative AI systems to detect vulnerabilities, biases, and data leakage. • Policy Contribution: Co-authored national guidelines (CNIL's AI doctrine) and European (EDPB) frameworks on AI robustness, security, and model confidentiality.
2020 - 2023	Postdoctoral Researcher, Lab. for Topology and Neuroscience, EPFL, Lausanne Developed state-of-the-art topological machine learning methods, including the Persformer (the first Transformer architecture for topological ML). Co-led productization of research into the open-source library giotto-deep in partnership with the Swiss start-up Giotto.Ai. Supervised and mentored over 10 graduate student research projects.
2018 - 2019	Entrepreneur of General Interest (Data Scientist), Ministry of Public Action and Accounts Within the framework of the Adler Project, managed the complete o-to-i product lifecycle (user research, design, implementation, and deployment) of PHOEBUS, a graph-based ML solution for detecting atypical financial transactions. Conducted feasibility studies on entity extraction using early BERT architectures to construct enterprise knowledge graphs from unstructured text.

Education

2016 - 2020	PhD in Computer Science and Artificial Intelligence – INRIA Saclay, École Polytechnique <i>Persistence and Sheaf Theory: from theory to applications</i>, supervised by Steve Oudot. Focus on Topological Data Analysis.
2015 - 2016	M.Sc. in Pure Mathematics, Imperial College (London)
2012 - 2016	Engineering Degree (Diplôme d'ingénieur), École Polytechnique

Skills

Technical	Deep Learning in Python (PyTorch, Hugging Face, Transformers), R , open-source library development, Git/GitHub.
AI Safety & Ops	LLM Evaluation, Adversarial Robustness, Red-teaming, Privacy-Preserving ML, Product Roadmap & Prioritization, Stakeholder Management.
Languages	French (Native), English (Professional, C1).

Key Publications and Talks

Publications

Singular Value Representation: A New Graph Perspective On Neural Networks

Accepted at AISTATS '23, written with my advisee Dan Meller. Analyzed structural properties of deep neural networks.

Persformer: A Transformer Architecture for Topological Machine Learning

Introduced the first Transformer-based architecture for topological data analysis.

Audit techniques for Large Language Models

Technical report outlining evaluation frameworks and safety auditing guidelines for LLMs (supervised Henri Duprieu's work).

Embrace the opaque: A sociology of the xAI community

Socio-technical study investigating explainability (xAI) and research-to-practice alignment.

Talks

Invited Speaker – INESIA Scientific Days

July 2023, Paris, France

Invited Speaker – ELSA Workshop: Privacy Preserving Machine Learning

March 2023, Bertinoro, Italy

Accepted Presentation – EASST'24 workshop: Sociotechnical dimensions of explainable and transparent AI.

July 2024, Amsterdam, Netherlands

Invited Speaker – ICLR'24 workshop: Privacy Regulation and Protection in Machine Learning

May 2024, Vienna, Austria